

Introduction to the Proper Orthogonal Decomposition (POD)

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Summary

- 1 Introduction to POD
- 2 Principle of Proper Orthogonal Decomposition (POD)
- 3 Initial Data for POD
- 4 Compute the spatial POD modes
- 5 Application
- 6 Conclusion

Introduction to POD

Key Advantages

- **Dimensionality Reduction**

POD extracts the most important modes from high-dimensional data, enabling efficient representation with fewer degrees of freedom.

- **Data-Driven Approach**

POD is purely data-driven and does not require prior knowledge of the governing equations.

- **Compatibility with Numerical Methods**

POD can be seamlessly integrated with numerical methods like FEM for reduced-order modeling (ROM).

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- **Noise Reduction**

By focusing on dominant modes, POD can filter out noise and less significant features in the data.

- **Real-Time Applications**

Reduced-order models based on POD enable real-time simulations and control.

- **Non-Intrusive Nature**

POD can be applied as a post-processing tool without modifying existing solvers or codes.

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Key Advantages

- **Versatility**

POD is applicable across various fields, including fluid dynamics, structural mechanics, and signal processing.

- **Foundation for Advanced Techniques**

POD serves as a basis for more advanced techniques like Dynamic Mode Decomposition (DMD).

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Principle of Proper Orthogonal Decomposition (POD)

Mathematical Formulation

A solution $\mathbf{u}(\mathbf{x}, t)$ is approximated as a linear combination of POD modes $\phi_i(\mathbf{x})$ and time-dependent coefficients $a_i(t)$:

$$\mathbf{u}(\mathbf{x}, t) \approx \sum_{i=1}^L a_i(t) \phi_i(\mathbf{x})$$

where :

- $\phi_i(\mathbf{x})$ are the spatial POD modes,
- $a_i(t)$ are the temporal coefficients,
- L is the number of retained modes.

Principle of Proper Orthogonal Decomposition (POD)

Objective

POD is a method for extracting dominant structures (modes) from high-dimensional data, enabling efficient dimensionality reduction.

Key Idea

The POD modes ϕ_i are computed to maximize the captured energy of the system, ensuring an optimal low-dimensional representation.

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Initial Data for POD

Data Requirements

POD requires a dataset that captures the spatio-temporal evolution of the system.

- The dataset should represent the system's behavior over time and space.
- Typically, this data is collected from simulations or experiments.

Concept of Snapshots

Definition

A **snapshot** is an instantaneous representation of the field at a specific time t_k .

- For a system described by a field $\mathbf{u}(\mathbf{x}, t)$, a snapshot is :

$$\mathbf{u}_k(\mathbf{x}) = \mathbf{u}(\mathbf{x}, t_k)$$

- $\mathbf{x}$: Spatial coordinates.
- t_k : Discrete time.
- M : Total number of snapshots.

Snapshot Matrix

Organization of Snapshots

Snapshots are organized into a matrix $\mathbf{X}$ of size $N \times M$.

$$\mathbf{X} = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \cdots \quad \mathbf{u}_M]$$

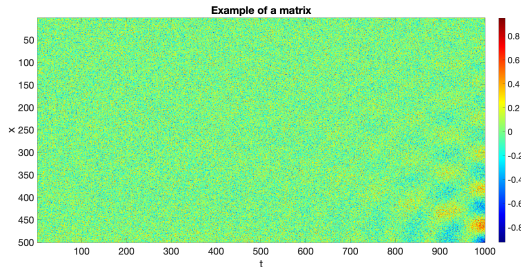
- Each column $\mathbf{u}_k$ represents a snapshot.
- N : Number of spatial points.
- M : Number of snapshots.

Snapshot Matrix

Organization of Snapshots

Snapshots are organized into a matrix $\mathbf{X}$ of size $N \times M$.

$$\mathbf{X} =$$



- Each column represents a snapshot.

Objective of POD

Main Goal

POD uses the snapshot matrix to extract dominant modes, maximizing the energy captured by each mode.

- The dominant modes provide a reduced-order representation of the system.
- This enables efficient simulations and analysis.

Objective of POD

Summary

- POD requires spatio-temporal data organized as snapshots.
- Snapshots are arranged in a matrix for mode extraction.
- The goal is to maximize energy capture for efficient modeling.

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Compute the Correlation Matrix

Correlation Matrix

The correlation matrix $\mathbf{C}$ is computed as :

$$\mathbf{C} = \frac{1}{\alpha} \mathbf{X} \mathbf{X}^T$$

- $\mathbf{C}$ is an $N \times N$ matrix.
- It captures the temporal correlations between the snapshots.

Note

This matrix is symmetric and positive semi-definite.

Solve the Eigenvalue Problem

Spatial POD Modes

Solve for the eigenvalues λ_i and eigenvectors ϕ_i of $\mathbf{C}$:

$$\mathbf{C}\phi_i = \lambda_i\phi_i$$

- λ_i represents the energy associated with the i -th mode.
- ϕ_i are the spatial modes. ϕ_i are orthogonal : $\phi_i^T \phi_j = \delta_{ij}$.

Note

The eigenvalues are assumed to be organized in descending order :
 $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_N$.

Determine the Temporal Coefficients

Temporal Coefficients

The temporal coefficients $a_i(t_k)$ are computed as the projection of the snapshots onto the POD modes :

$$a_i(t_k) = \phi_i^T \mathbf{u}_k \Rightarrow \mathbf{a} = \phi^T \mathbf{X}$$

- $a_i(t_k)$ represent the time evolution of the i^{th} mode at time t_k .

Note

These coefficients describe how the spatial modes vary over time.

Summary of the POD Process

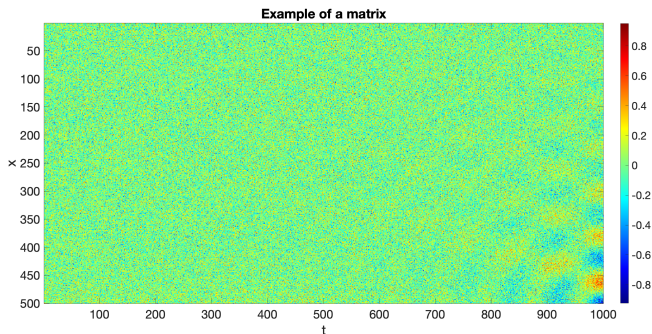
Steps to Calculate POD Modes

- 1 Construct the snapshot matrix $\mathbf{X}$.
- 2 Compute the correlation matrix $\mathbf{C} = \frac{1}{\alpha} \mathbf{X} \mathbf{X}^T$.
- 3 Compute the spatial POD modes $\mathbf{C} \phi_i = \lambda_i \phi_i$.
- 4 Determine the temporal coefficients $a_i(t_k) = \phi_i^T \mathbf{u}_k$.

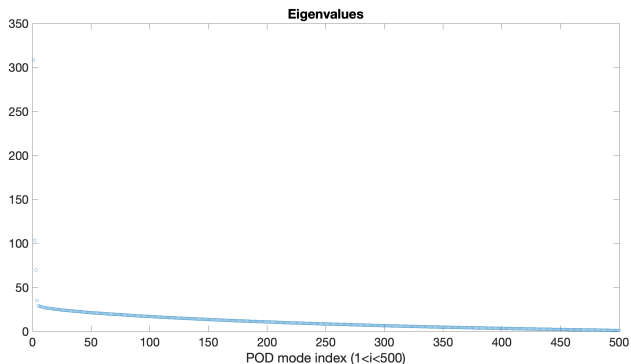
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Plot of the matrix $\mathbf{X}$ containing the snapshots $\mathbf{u}(\mathbf{x}, t)$



Plot of the eigenvalues $\lambda_{1 \leq i \leq N}$ of the correlation matrix $\mathbf{C}$



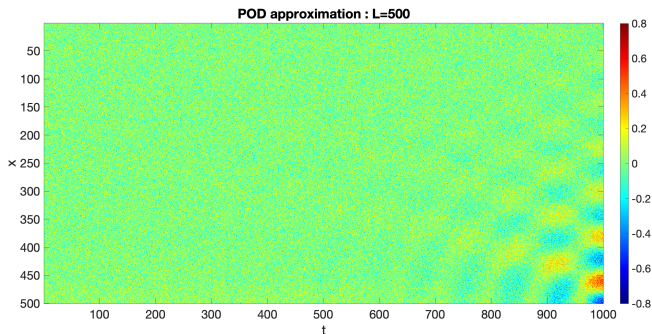
POD description

Approximation

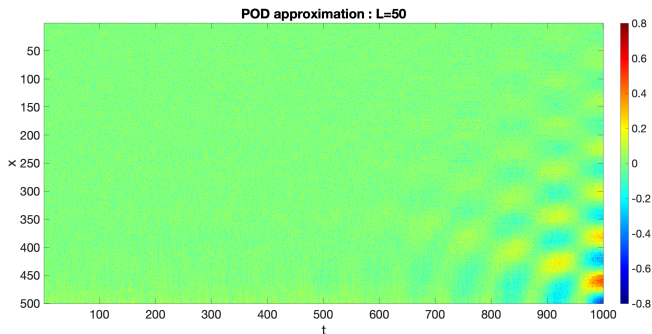
The data matrix $\mathbf{X}$ is approximated as a linear combination of POD modes $\phi_i(\mathbf{x})$ and time-dependent coefficients $a_i(t)$:

$$\mathbf{u}(\mathbf{x}, t) \approx \sum_{i=1}^{L \leq N} a_i(t) \phi_i(\mathbf{x})$$

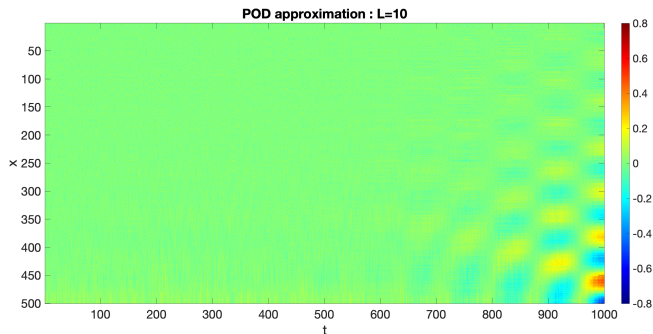
Plot of the POD approximation : $L=500$



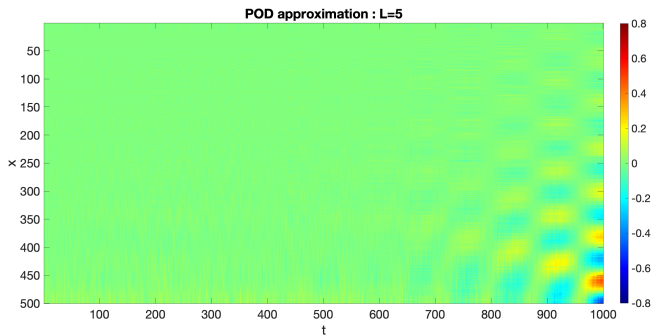
Plot of the POD approximation : $L=50$



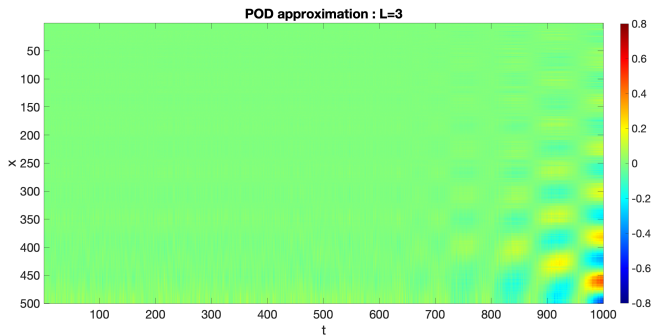
Plot of the POD approximation : $L=10$



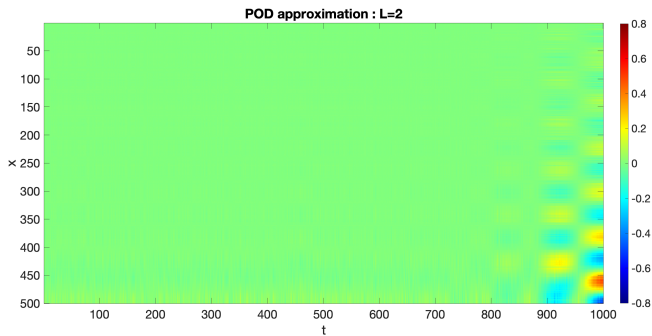
Plot of the POD approximation : $L=5$



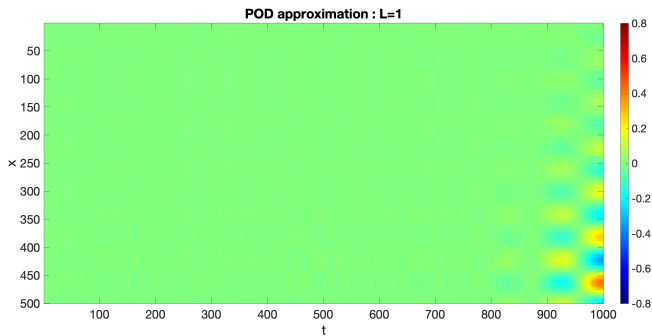
Plot of the POD approximation : $L=3$



Plot of the POD approximation : $L=2$



Plot of the POD approximation : $L=1$



Data storage

Memory size

- Matrix $\mathbf{X}$: $N \times M = 500\,000$
- POD approximation : $L \times (M + N) = 7\,500$ (L=5)

Interests of the POD approximation

- Gain of 98,5% of the memory space for L=5
- Improvement of data quality by filtering out the noise

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Conclusion

Key Takeaways

- POD modes are computed from snapshots to reduce the dimensionality of the system.
- The process involves constructing a snapshot matrix, computing a correlation matrix, and solving an eigenvalue problem.
- The resulting spatial modes and temporal coefficients provide a compact representation of the system.

Thank you !

Questions ?